

Genome Sequence of a *Clostridioides difficile* Strain Isolated from Feces from a Patient in Southern Thailand.

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Clostridioides difficile is a Gram-positive, obligate anaerobic, toxin-producing bacillus that is linked to antibiotic-associated diarrhea. Here, we report the whole-genome sequence of a *C. difficile* strain isolated from stool from a patient, using next-generation sequencing (MGISEG-2000). De novo assembly revealed a genome length of 4,208,266 bp. Multilocus sequence typing (MLST) results showed that the isolate belonged to sequence type 23 (ST23).

Evaluation of functionality of type II toxin-antitoxin systems of *Clostridioides difficile* R20291.

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Clostridioides difficile is a nosocomial, Gram-positive, strictly anaerobic, spore-forming pathogen capable of colonizing and proliferating in the human intestine. In bacteria, it has been shown that the Toxin-Antitoxin systems mediate the cellular response to external stress by initiating processes such as biofilm formation and programmed cell death. This work aims to evaluate the functionality of four type II TA modules of *Clostridioides difficile* R20291. We performed bioinformatic analysis to search for putative TA systems using the TADB platform. Then we performed a heterologous expression assay to evaluate the functionality of these systems. Our results showed that the MazEF and RelBE systems were functional, suggesting that their corresponding toxins possess an endoribonuclease activity. In conclusion, MazEF and RelBE systems of *C. difficile* R20291 are functional in a heterologous expression system.

Septic arthritis post anterior cruciate ligament reconstruction due to *Clostridioides difficile*.

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Clostridioides difficile is an obligate anaerobe ubiquitous in the environment and is of particular interest in the healthcare setting as a cause of healthcare associated infection usually presenting with colitis. Extracolonic manifestations of *C. difficile* infection are less common with only rare reports of septic arthritis primarily in the setting of relative or overt immunocompromise. This report details the case of a 31-year-old immunocompetent male presenting with clinical features of septic arthritis, three weeks post right knee anterior cruciate ligament (ACL) reconstruction using a native hamstring tendon graft. *C. difficile* was isolated from two different samples of the synovial tissue from a subsequent arthroscopic washout and synovectomy. The ACL graft was retained. The isolate underwent whole genome sequencing and was found to be *tcdA* and *tcdB* gene deficient. Susceptibility testing showed susceptibility to benzylpenicillin and metronidazole. The patient received a two-week course of intravenous benzylpenicillin and four weeks of oral metronidazole. At one-year post cessation of antibiotics the patient has no clinical evidence of recurrence. This is the first known reported case of *C. difficile* septic arthritis in an immunocompetent patient. It demonstrates successful treatment of post-ACL septic arthritis with a graft retention strategy.

Identification and preclinical evaluation of MMV676558 as a promising therapeutic candidate against *Clostridioides difficile*.

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Clostridioides difficile, a gram-positive, toxin-producing, spore-forming anaerobe, is a major cause of antibiotic-associated diarrhoea. The bacterium's intrinsic drug resistance limits current treatment options to fidaxomicin and vancomycin for initial episodes, with anti-toxin B monoclonal antibody or faecal microbiota transplantation recommended for complicated or recurrent cases. This underscores the urgent need for novel therapeutics. In this study, we screened the MMV Pathogen Box at a 10 μ M concentration against *C. difficile* R20291. Primary hits were evaluated for minimum inhibitory concentrations (MIC), killing kinetics, and biofilm inhibition. Bacterial cytological profiling (BCP) and transmission electron microscopy (TEM) were employed to study the mode of action. MMV676558 was further tested in a mouse model to assess survival, histopathology, and gut microbiota effects. We identified nineteen hits that inhibited over 50 % of *C. difficile* growth. MIC assays revealed three hits with MICs below 16 μ g/mL: MMV676558, MMV688755, and MMV690027. These hits were effective against various *C. difficile* ribotypes. Killing kinetics were comparable or superior to vancomycin and fidaxomicin, and biofilm assays showed inhibitory effects. BCP and TEM analyses suggested membrane function disruption as the mode of action. Furthermore, MMV676558 demonstrated a protective effect in mice, with favourable histopathology and gut microbiota profiles. Given the urgent threat posed by *C. difficile* antibiotic resistance, discovering new treatments is a top priority. Our study identified three promising hits from the MMV Pathogen Box, with MMV676558 showing significant in vivo potential for further evaluation.

Functional characterization of Fur from the strict anaerobe *Clostridioides difficile* provides insight into its redox-driven regulatory capacity.

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Clostridioides (formerly *Clostridium*) *difficile* is a leading cause of infectious diarrhea associated with antibiotic therapy. The ability of this anaerobic pathogen to acquire enough iron to proliferate under iron limitation conditions imposed by the host largely determines its pathogenicity. However, since high intracellular iron catalyzes formation of deleterious reactive hydroxyl radicals, iron uptake is tightly regulated at the transcriptional level by the ferric uptake regulator Fur. Several studies relate lacking a functional fur gene in *C. difficile* cells to higher oxidative stress sensitivity, colonization defect and less toxigenicity, although Fur does not appear to directly regulate either oxidative stress response genes or pathogenesis genes. In this work, we report the functional characterization of *C. difficile* Fur and describe an additional oxidation sensing Fur-mediated mechanism independent of iron, which affects Fur DNA-binding. Using electrophoretic mobility shift assays, we show that Fur binding to the promoters of fur, feoA and fldX genes, identified as iron and Fur-regulated genes in vivo, is specific and does not require co-regulator metal under reducing conditions. Fur treatment with H₂O₂ produces dose-dependent soluble high molecular weight species unable to bind to target promoters. Moreover, Fur oligomers are dithiotreitol sensitive, highlighting the importance of some interchain disulfide bond(s) for Fur oligomerization, and hence for activity. Additionally, the physiological electron transport chain NADPH-thioredoxin reductase/thioredoxin from *Escherichia coli* reduces inactive oligomerized *C. difficile* Fur that recovers activity. In conjunction with available transcriptomic data, these results suggest a previously underappreciated complexity in the control of some members of the Fur regulon that is based

on Fur redox properties and might be fundamental for the adaptive response of *C. difficile* during infection.

Approaching toxigenic *Clostridia* from a One Health perspective.

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Spore-forming pathogens have a unique capacity to thrive in diverse environments, and with temporal persistence afforded through their ability to sporulate. Their prevalence in diverse ecosystems requires a One Health approach to identify critical reservoirs and outbreak-associated transmission chains, given their capacity to freely move across soils, waterways, foodstuffs and as commensals or infecting pathogens in human and animal populations. Among anaerobic spore-formers, genomic resources for pathogens including *C. botulinum*, *C. difficile*, and *C. perfringens* enable our capacity to identify common and unique factors that support their persistence in diverse reservoirs and capacity to cause disease. Publicly available genomic resources for spore-forming pathogens at NCBI's Pathogen Detection program aid outbreak investigations and longitudinal monitoring in national and international programs in public health and food safety, as well as for local healthcare systems. These tools also enable research to derive new knowledge regarding disease pathogenesis, and to inform strategies in disease prevention and treatment. As global community resources, the continued sharing of strain genomic data and phenotypes further enhances international resources and means to develop impactful applications. We present examples showing use of these resources in surveillance, including capacity to assess linkages among clinical, environmental, and foodborne reservoirs and to further research investigations into factors promoting their persistence and virulence in different settings.

Characterization of a stable, metronidazole-resistant *Clostridium difficile* clinical isolate.

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Clostridium difficile are gram-positive, spore forming anaerobic bacteria that are the leading cause of healthcare-associated diarrhea, usually associated with antibiotic usage. Metronidazole is currently the first-line treatment for mild to moderate *C. difficile* diarrhea however recurrence occurs at rates of 15-35%. There are few reports of *C. difficile* metronidazole resistance in the literature, and when observed, the phenotype has been transient and lost after storage or exposure of the bacteria to freeze/thaw cycles. Owing to the unstable nature of the resistance phenotype in the laboratory, clinical significance and understanding of the resistance mechanisms is lacking. Genotypic and phenotypic characterization was performed on a metronidazole resistant clinical isolate of *C. difficile*. Whole-genome sequencing was used to identify potential genetic contributions to the phenotypic variation observed with molecular and bacteriological techniques. Phenotypic observations of the metronidazole resistant strain revealed aberrant growth in broth and elongated cell morphology relative to a metronidazole-susceptible, wild type NAP1 strain. Comparative genomic analysis revealed single nucleotide polymorphism (SNP) level variation within genes affecting core metabolic pathways such as electron transport, iron utilization and energy production. This is the first characterization of stable, metronidazole resistance in a *C. difficile* isolate. The study provides an in-depth genomic and phenotypic analysis of this strain and provides a foundation for future studies to elucidate mechanisms conferring metronidazole resistance in *C. difficile* that have not been previously described.

1H, 13C and 15N resonance assignments and structure prediction of translation initiation factor 1 from Clostridium difficile.

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Clostridium difficile is a gram-positive, toxin-producing, anaerobic bacterium whose virulence factors and mechanisms of pathogenesis require further investigation. *C. difficile* infections (CDI) result in the severe and potentially fatal gastrointestinal diseases pseudomembranous colitis and toxic megacolon following extensive broad spectrum antibiotic treatment. The increasing *C. difficile* fatalities are a result of the bacteria's growing antibiotic resistance and consequential CDI recurrence, which led to the unmet need for new CDI treatment. Bacterial protein synthesis is an essential metabolic process and an effective target for antibacterial agents. Translation initiation factor 1 from *C. difficile* (Cd-IF1) is the smallest of the three initiation factors that acts to establish the 30S initiation complex to initiate translation during protein biosynthesis. Here we report the complete NMR 1H, 13C and 15N chemical shift assignments of Cd-IF1 as the basis for NMR structure determination and interaction studies. Secondary structure analyses have identified five β -strands and one short α -helix arranged in the sequential order β 1- β 2- β 3- α 1- β 4- β 5, which is supported by 15N-{1H} heteroNOEs. The assigned chemical shifts were used to conduct structure prediction by CS-Rosetta. The predicted structure suggests that Cd-IF1 adopts the typical β -barrel structure and is composed of an oligomer-binding motif.

Elective Stoma Reversal Has a Higher Incidence of Postoperative Clostridium Difficile Infection Compared With Elective Colectomy: An Analysis Using the American College of Surgeons National Surgical Quality Improvement Program and Targeted Colectomy Databases.

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Clostridium difficile infection is caused by the proliferation of a gram-positive anaerobic bacteria after medical or surgical intervention and can result in toxic complications, emergent surgery, and death. This analysis evaluates the incidence of *C. difficile* infection in elective restoration of intestinal continuity compared with elective colon resection. This was a retrospective database review of the 2015 American College of Surgeons National Surgical Quality Improvement Project and targeted colectomy database. The intervention cohort was defined as the primary Current Procedural Terminology codes for ileostomy/colostomy reversal (44227, 44620, 44625, and 44626) and International Classification of Diseases codes for ileostomy/colostomy status (VV44.2, VV44.3, VV55.2, VV55.3, Z93.2, Z93.3, Z43.3, and Z43.2). A total of 2235 patients underwent elective stoma reversal compared with 10403 patients who underwent elective colon resection. Multivariate regression modeling of the impact of stoma reversal on postoperative *C. difficile* infection risk was used as the study intervention. The incidence of *C. difficile* infection in the 30 days after surgery was measured. The incidence of *C. difficile* infection in the 30-day postoperative period was significantly higher (3.04% vs 1.25%; $p < 0.001$) in patients undergoing stoma reversal. After controlling for differences in cohorts, regression analysis suggested that stoma reversal (OR = 2.701 (95% CI, 1.966-3.711); $p < 0.001$), smoking (OR = 1.520 (95% CI, 1.063-2.174); $p = 0.022$), steroids (OR = 1.677 (95% CI, 1.005-2.779); $p = 0.048$), and disseminated cancer (OR = 2.312 (95% CI, 1.437-3.719); $p = 0.001$) were associated with *C. difficile* infection incidence in the 30-day postoperative period. The study was limited because it was a retrospective database review with observational bias.

Patients who undergo elective stoma reversal have a higher incidence of postoperative C difficile infection compared with patients who undergo an elective colectomy. Given the impact of postoperative C difficile infection, a heightened sense of suspicion should be given to symptomatic patients after stoma reversal. See at Video Abstract at <http://links.lww.com/DCR/A553>.

Updates on Clostridium difficile spore biology.

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Clostridium difficile is a Gram-positive, anaerobic spore former, and an important nosocomial pathogenic bacterium. C. difficile spores are the morphotype of transmission and recurrence of the disease. The formation of C. difficile spores and their subsequent germination are essential processes during the infection. Recent in vitro and in vivo work has shed light on how spores are formed and the timing of in vivo sporulation in a mouse model. Advances have also been made in our understanding of the machineries involved in spore germination, and how antibiotic-induced dysbiosis affects the metabolism of bile salts and thus impacts C. difficile germination in vivo. Studies have also attempted to identify how C. difficile spores interact with the host's intestinal mucosa. Spore resistance has also been revisited by several groups highlighting the extreme resistance of this morphotype to traditional food processing regimes and disinfectants used in clinical settings. Therefore, the aim of this review is to summarize recent advances on spore formation/germination in vitro and in vivo, spore-host interactions, and spore resistance that contribute to our knowledge of the role of C. difficile spores in the infectious process.
